



Automatic Street Light: Solar & Grid Hybrid

A smart lighting solution that seamlessly switches between solar power and grid power — ensuring uninterrupted, energy-efficient illumination for urban and rural roads.

SMART INFRASTRUCTURE

SUSTAINABLE ENERGY



The Challenge

Why Traditional Street Lights Fall Short

Energy Wastage

Conventional lights run on grid power around the clock, consuming significant electricity even during daylight hours.

Power Outages

Grid failures leave streets dark, creating safety hazards for pedestrians and motorists alike.

High Operational Cost

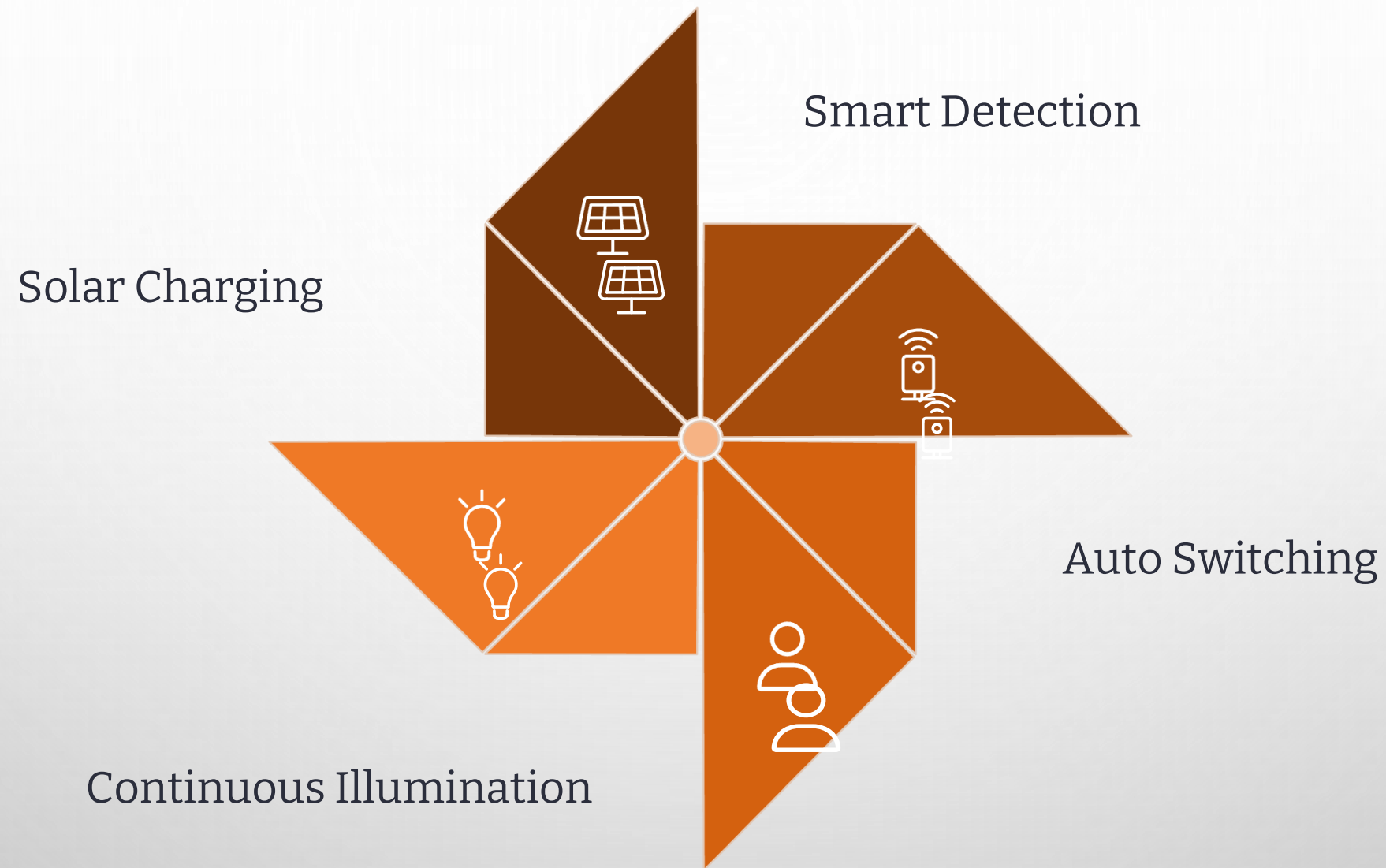
Rising electricity tariffs make large-scale street lighting a significant burden on municipal budgets.



Our Proposed Solution

An **automatic dual-power street light** that intelligently switches between solar energy and grid electricity based on solar availability, battery status, and ambient light levels — delivering continuous, reliable illumination at reduced cost.

How It Works



The system's embedded controller monitors battery voltage and ambient light continuously, triggering a seamless switchover to grid power only when solar reserves fall below a safe threshold — ensuring zero interruption in street lighting.

Hardware Components



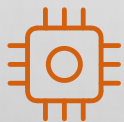
Solar Panel

Converts sunlight to DC electricity, charging the onboard battery during daylight hours for night-time operation.



Rechargeable Battery

Stores harvested solar energy; powers the LED during night. Battery level is continuously monitored for smart switching.



Microcontroller

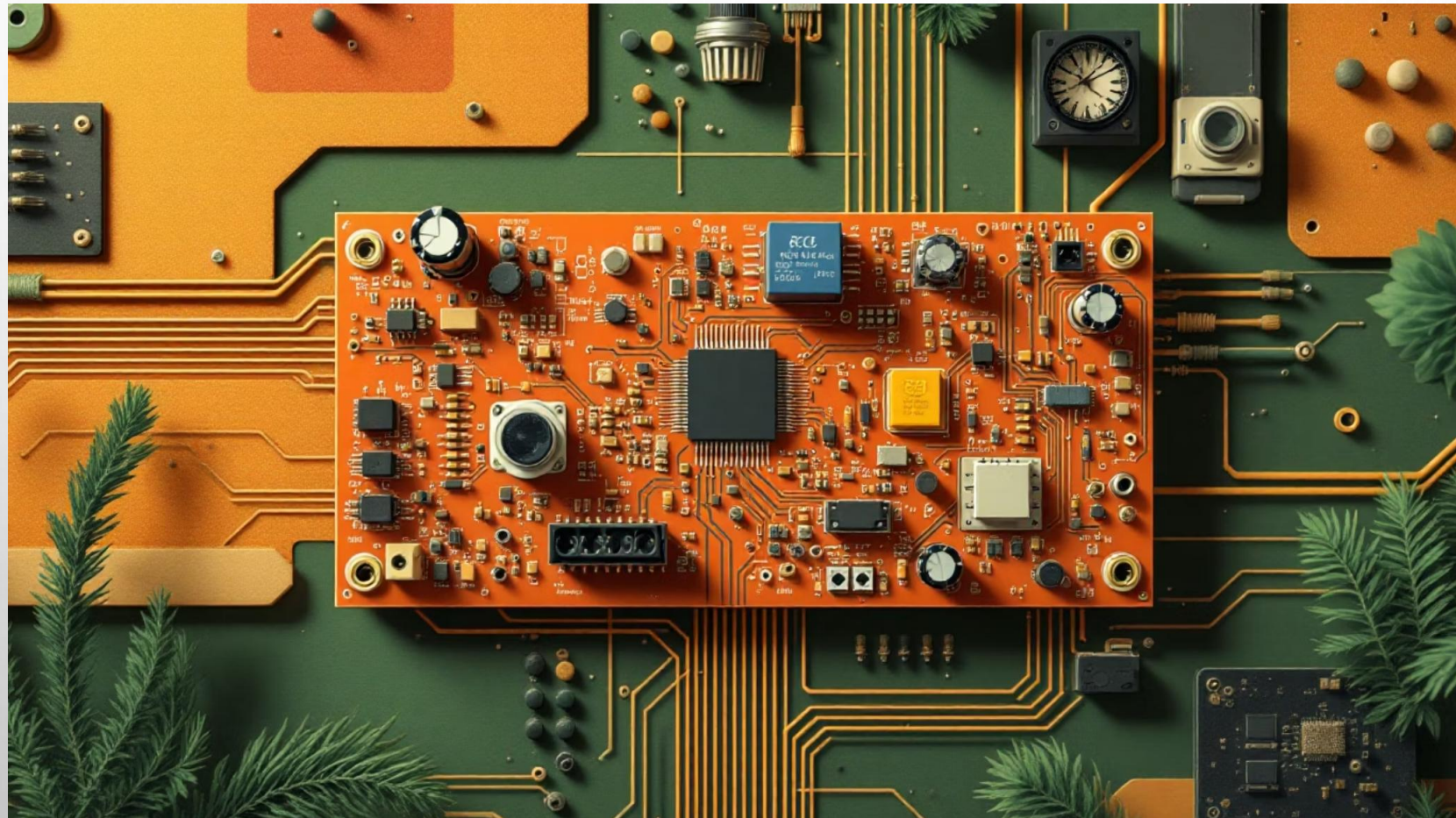
The brain of the system — reads sensor data and controls relay switching between solar and grid power automatically.



LDR Sensor

Detects ambient light levels to trigger automatic ON/OFF operation — eliminating manual switching entirely.

Tech Stack



Embedded C / Arduino IDE

Firmware for sensor reading, decision logic, and relay control.

Relay Module

Electronically switches load between solar battery output and AC grid supply safely.

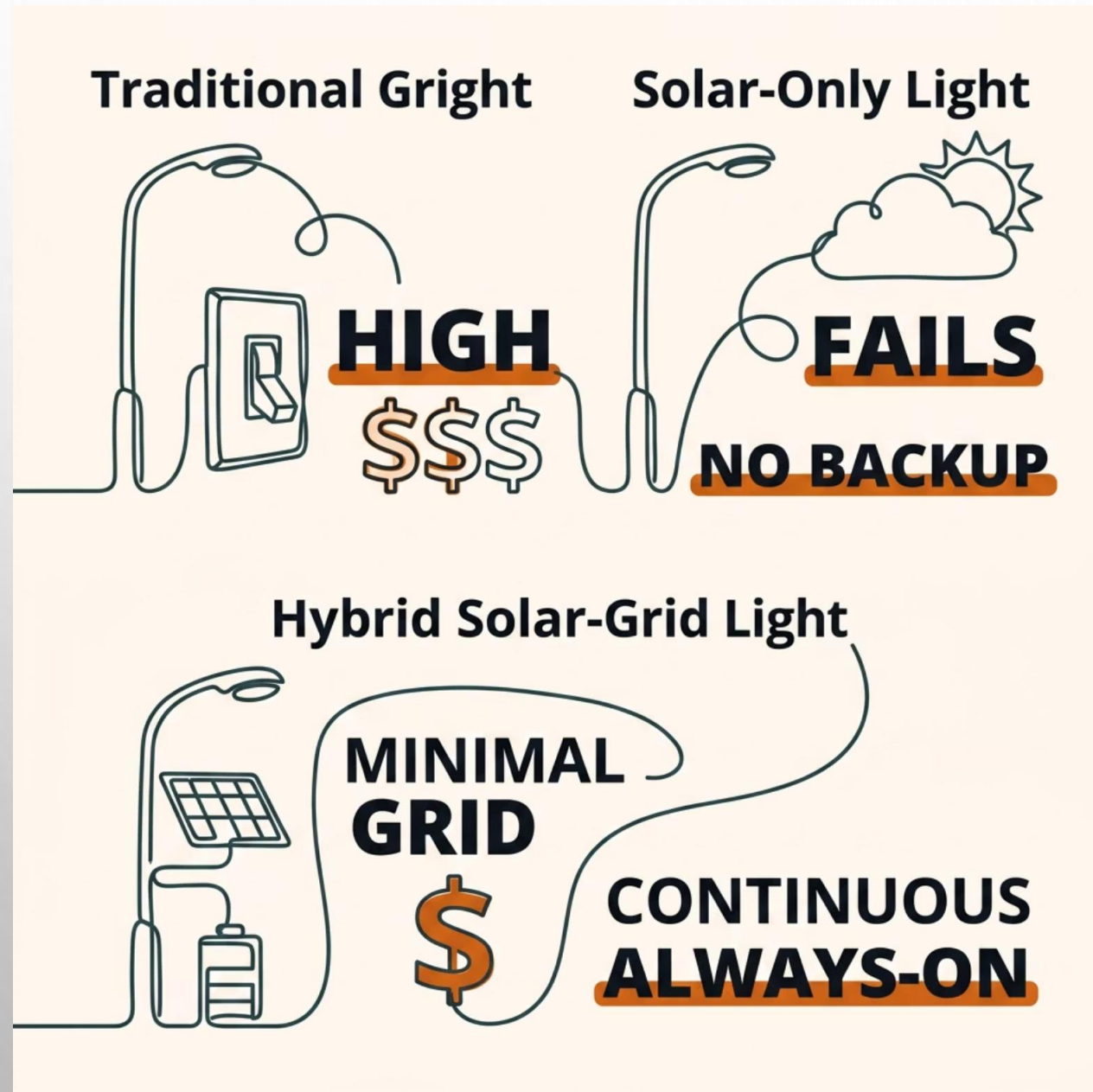
LED Driver Circuit

Ensures constant current to the LED for consistent brightness across power sources.

Voltage Regulator

Stabilises supply to the microcontroller regardless of input source fluctuations.

Feasibility & Cost Effectiveness



Why This Makes Financial Sense

The hybrid model reduces grid electricity consumption by up to **70%** compared to conventional street lights. The initial hardware investment is recovered within **2-3 years** through electricity savings, with a system lifespan exceeding **10 years**.

- Low maintenance, widely available components, and compatibility with existing pole infrastructure make deployment straightforward for municipalities.



Sustainability Impact

70%

Grid Energy Saved

Reduction in electricity drawn from the grid per unit annually.

0

Manual Intervention

Fully automated ON/OFF and power-source switching — zero operator effort.

24/7

Uptime Guaranteed

Continuous lighting even during grid outages, powered by stored solar energy.

Future Impact & Scalability

1

Smart City Integration

IoT-enabled variants can report real-time energy data to municipal dashboards for centralised monitoring.

2

Motion-Adaptive Dimming

Next-gen versions can dim to 30% when no movement is detected, saving additional energy.

3

Rural Electrification

Ideal for areas with unreliable grid supply — providing safe, autonomous street lighting without infrastructure investment.

